Part I
Cryptography-based communications

1 IPv6

Why IPv6?

To solve IPv4 limitations

- IPv4 address classes are overused
- 32 bits for IPv4 address
- Almost no security on IPv4 design

IPv6: some characteristics

- 1 interface $\Leftrightarrow$ as much addresses as necessary
- Lifetime defined for each address
- Format:
  - *IPv6 prefix* = *Global Routing Prefix* on 48 bits + *Subnet ID* on 16 bits
  - *Interface ID* on 64 bits
- Types:
  1. **unicast**
  2. **anycast**
  3. **multicast**

Address types

Unicast

- Hexadecimal 16 bits-blocks
- Successive 0-blocs can be omitted
- Loopback : $\rightarrow$ :: 1
- 0.0.0.0 $\rightarrow$ ::

Hierarchical decomposition

- $11 : 22 : 33 : ab : 44 : 55 : 66 : 77$

$\rightarrow$ $11 : 22 : 33 : ab : /64$ (*Subnet ID*)

$\rightarrow$ $11 : 22 : 33$ (*Global Routing Prefix*)

Address types

Prefixes

- **IPv6-address/prefix-length-in-bits**
- Prefix for *local link* = $fe80 :: /10$

→ built from MAC address

⇒ single address for local network

Multicast

- Group interfaces with a common characteristic (prefix = $ff00 :: /8$)
- Range limited to the node, the segment or the Internet
- Multicast ≠ broadcast IPv4

- Use a service without knowing a specific server

ICMPv6

- Extends classic ICMP

→ replaces *Internet Group Message Protocol*

→ replaces *Address Resolution Protocol*

- Address auto-configuration by discovering
  - neighbours
  - prefixes already used and duplicated addresses
  - physical link characteristics

Transition IPv4 - IPv6

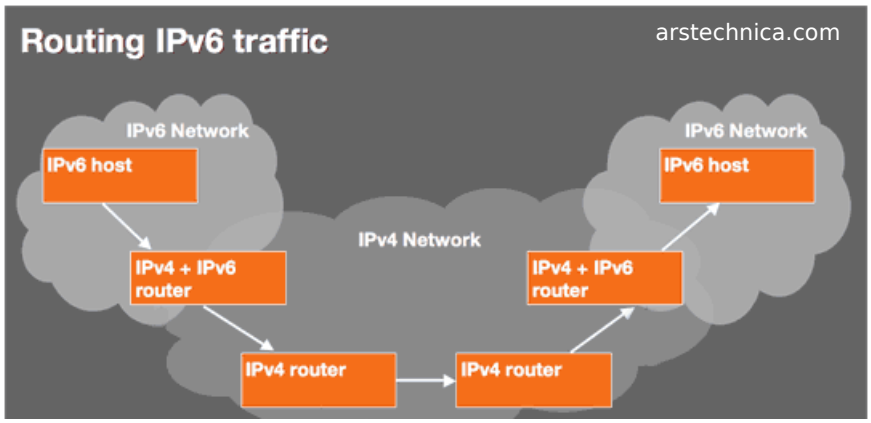
Slow adoption

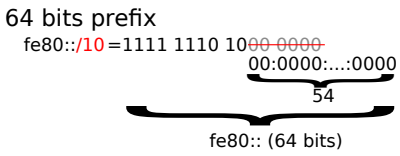
→ easy adaptation from IPv4 to IPv6

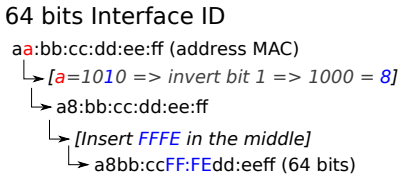
⇒ IPv4 is supported by default on IPv6 software

Protocol 6to4 (RFC 3056)

- Use IPv4 to transport IPv6
- *Tunnels 6to4* slice IPv6 packets → encapsulation into IPv4 paquets and conversely







IPv6 routing

StateLESS @ auto-configuration

Local address

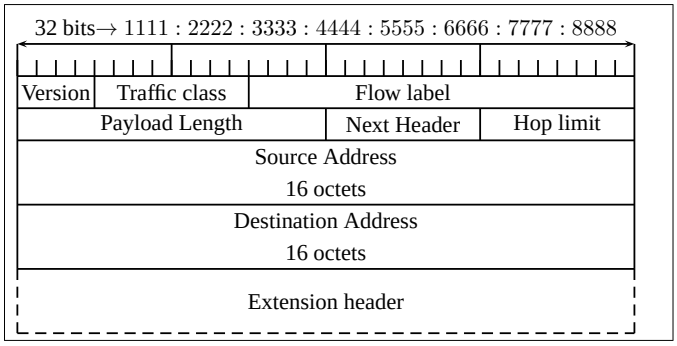
Address validation

1. ICMPv6 *Neighbor Solicitation* request
2. ICMPv6 *Router Advertisement* answer $\Leftrightarrow$ @ is already being used $\Rightarrow$ new @ generation
3. Test OK $\rightarrow$ looking for routers: either wait for *Router Advertisement* ; or send *Router Solicitation*

StateFUL @ auto-configuration

- DHCPv6 manages everything (DHCPv6 router may be located on a $\neq$ segment)
  1. Client: multicast request on the same segment or on the network
  2. Server gives the @ to use
  3. Client requests for @
- DHCPv6 servers may provide extra information

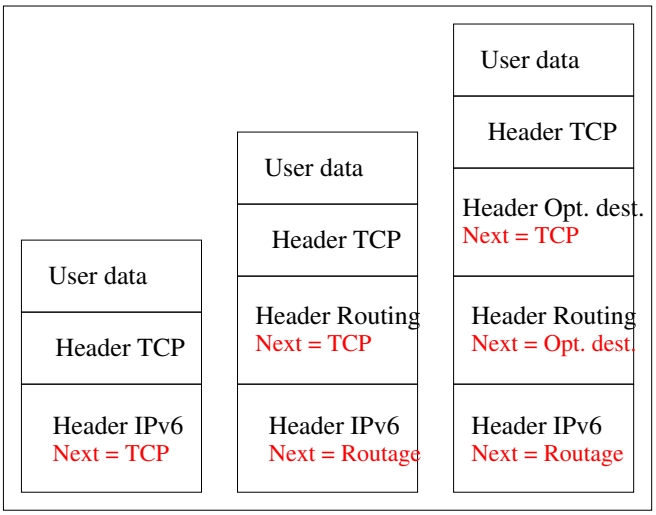
Datagram header



Extension headers

- Hop by hop: information for all routers on the path
- Routing: routers to follow
- Fragment: yes/no
- Authentication Header: yes/no
- Encapsulating Security Payload: yes/no
- Destination options: information for dest. only
- Upper layer (TCP, UDP...)

Extensions



```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
| Next Header   |  Payload Len  |          RESERVED             |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                 Security Parameters Index (SPI)               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      Sequence Number Field                    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                                                               |
+               Integrity Check Value-ICV (variable)            |
|                                                               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

```
                    BEFORE APPLYING AH
               ----------------------------------------
      IPv6  |              | ext hdrs |     |      |
            | orig IP hdr |if present| TCP | Data |
               ----------------------------------------
```

Mode transport

```
               ----------------------------------------------------------
      IPv6  |              |hop-by-hop, dest*, |     | dest |     |     |
            |orig IP hdr   |routing, fragment. | AH | opt* | TCP | Data |
               ----------------------------------------------------------
            |<--- mutable field processing -->|<-- immutable fields -->|
            |<---- authenticated except for mutable fields ---------->|

                 * = if present, could be before AH, after AH, or both
```

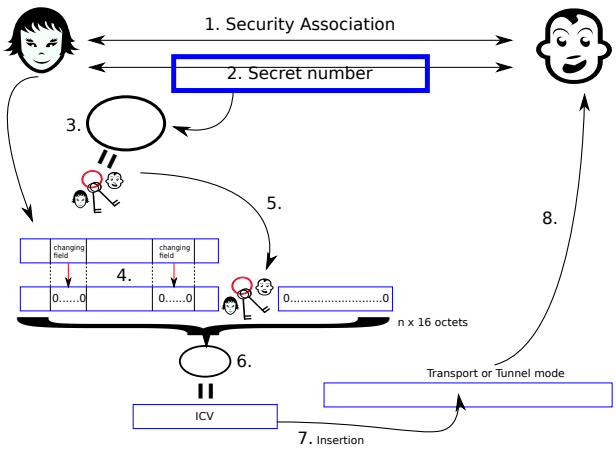
Mode tunnel

```
               ----------------------------------------------------------------
      IPv6  |              | ext hdrs*|    |              | ext hdrs*|    |    |
            |new IP hdr*|if present| AH |orig IP hdr*|if present|TCP|Data|
               ----------------------------------------------------------------
            |<--- mutable field -->|<-------- immutable fields -------->|
            |       processing     |
            |<-- authenticated except for mutable fields in new IP hdr ->|
```

Authentication Header: auth. and integrity

Format RFC2402bis-11

Datagram processing



Encapsulating Security Payload header: AH + encryption

2 IPsec

IPsec

Best location to manage Internet security?

1. *Application* layer?

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|               Security Parameters Index (SPI)                 |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      Sequence Number                          |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Payload Data* (variable)                   |
~                                                               ~
|                                                               |
+               +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|               |     Padding (0-255 bytes)                     |
+-+-+-+-+-+-+-+-+               +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                               |  Pad Length   | Next Header   |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|         Integrity Check Value-ICV   (variable)                |
~                                                               ~
|                                                               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Mode transport

```
            BEFORE APPLYING ESP
      ----------------------------------------
IPv6  |             | ext hdrs |     |      |
      | orig IP hdr |if present| TCP | Data |
      ----------------------------------------

            AFTER APPLYING ESP
      -------------------------------------------------------
IPv6  | orig |hop-by-hop,dest*.|   |dest|   |    | ESP   | ESP|
      |IP hdr|routing,fragment.|ESP|opt*|TCP|Data|Trailer| ICV|
      -------------------------------------------------------
                                |<--- encryption ---->|
                            |<------ integrity ------>|

   * = if present, could be before ESP, after ESP, or both
```

Mode tunnel

```
            BEFORE APPLYING ESP
      ----------------------------------------
IPv6  |             | ext hdrs |     |      |
      | orig IP hdr |if present| TCP | Data |
      ----------------------------------------

            AFTER APPLYING ESP

      -------------------------------------------------------------
IPv6  | new* |new ext |    | orig*|orig ext |    |    | ESP   | ESP|
      |IP hdr| hdrs*  |ESP|IP hdr| hdrs *  |TCP|Data|Trailer| ICV|
      -------------------------------------------------------------
                          |<--------- encryption ---------->|
                      |<------------ integrity ------------>|
   *= if present, depend of Security Architecture adopted
```

2. *Network* layer?

→ Users do not mind about security

IP security (Network)
Security services, cryptography algorithms and granularity levels

IPSec inside OSI

Application OSI layers 5, 6, 7
TCP/UDP OSI layers 4
IP / IPSEC OSI layer 3
Data Link and Physical layers OSI layers 1, 2

Security inside IPsec

- Connection = Security Association
- Security Policy Database
- Security Association Database
- Internet Key Exchange
- Authentication Header
- Encapsulating Security Payload

How IPsec works

Tunnel mode

1. Network A: IP packet $\rightarrow$ Gateway A
2. Gateway A: encapsulates and encrypts with IPsec $\rightarrow$ Gateway B
3. Gateway B: decrypts and decapsulates $\rightarrow$ Network B

Transport mode
IPsec header = original IP header + payload encryption

3 Secure Socket Layer / Transport Layer Security

SSL / TLS

Asymmetric encryption

- Protocol-independent
  1. client/server parameters negociation
  2. mutual authentication
  3. secret communication
  4. data integrity

SSL layers

SSL Record Layer
Manages encryption, fragmentation and compression

Layer 2

- SSL Handshake Protocol
- SSL Change Cipher
- SSL Alert

SSL connection

Alice				Bob
1	$\rightarrow$	SSL version, preferences, E_A	$\rightarrow$	
	$\leftarrow$	E_A(SSL version, choice, E_B)	$\leftarrow$	2
	$\leftarrow$	Certificate chain X. 509	$\leftarrow$	3
	$\leftarrow$	OK	$\leftarrow$	4
5	$\rightarrow$	$E_B(K_{\text{session}})$	$\rightarrow$	
6	$\rightarrow$	"Change cypher"	$\rightarrow$	
7	$\rightarrow$	OK	$\rightarrow$	
	$\leftarrow$	"Change cypher"	$\leftarrow$	8
	$\leftarrow$	OK	$\leftarrow$	9

SSL Record

Protocol steps

1. Fragmentation in fixed-size blocks
2. Compression optional
3. Concatenation (sender's secret key + message number + length + compressed block) $\rightarrow$ hashing for authentication/integrity
4. Encryption with K_{session} (concatenation + hashing result)
5. SSL header is created and sent with TCP